

ASSIGNMENT NO. 10: Window Functions and Analytical Queries

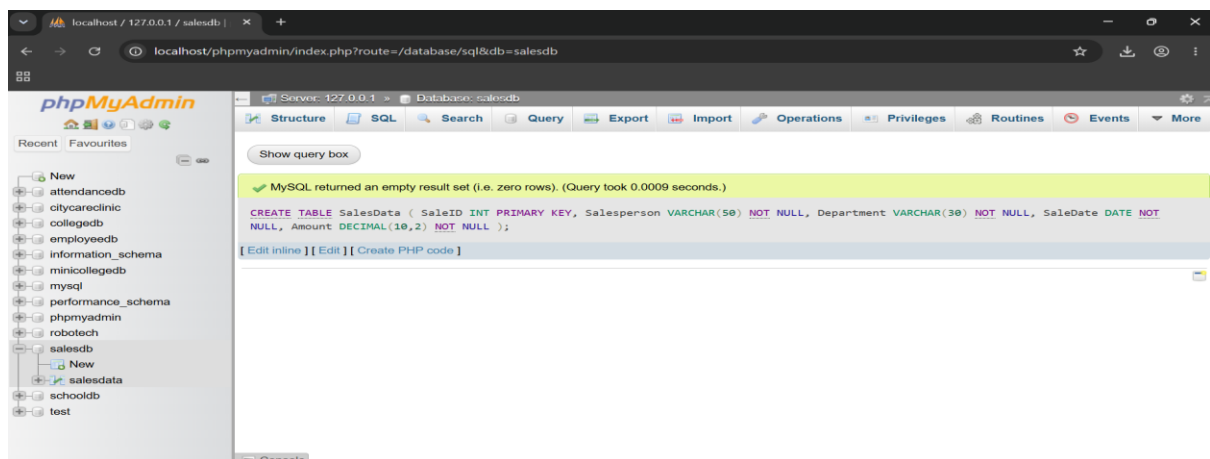
Q1) Create a MySQL database named SalesDB and a table named SalesData. SalesData Table

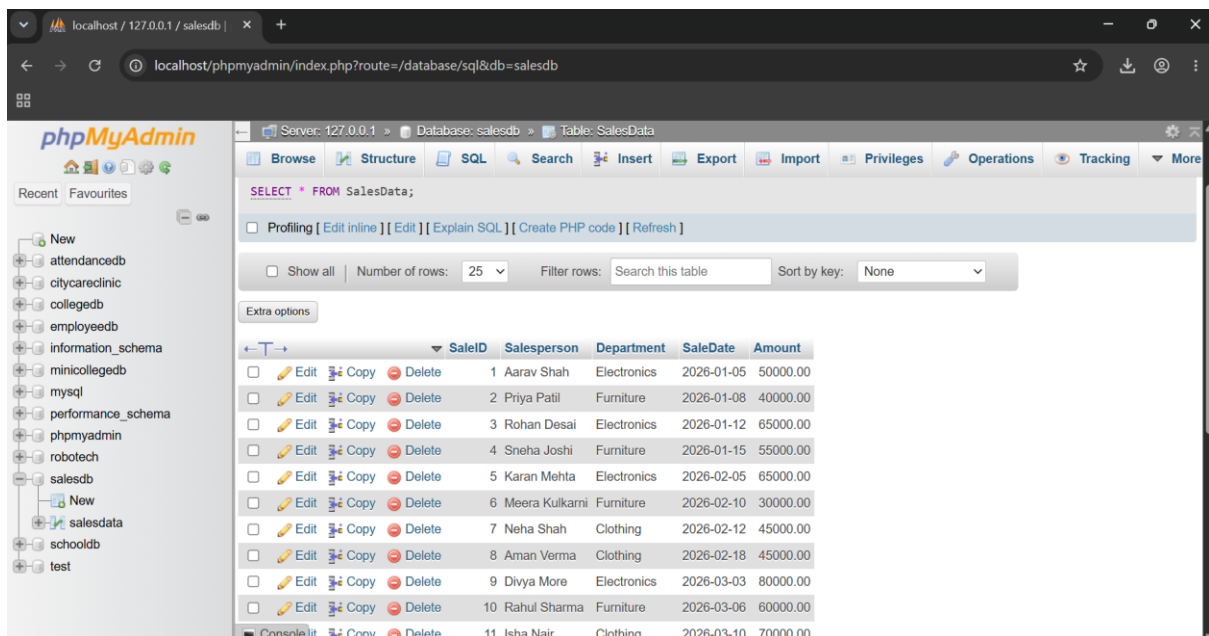
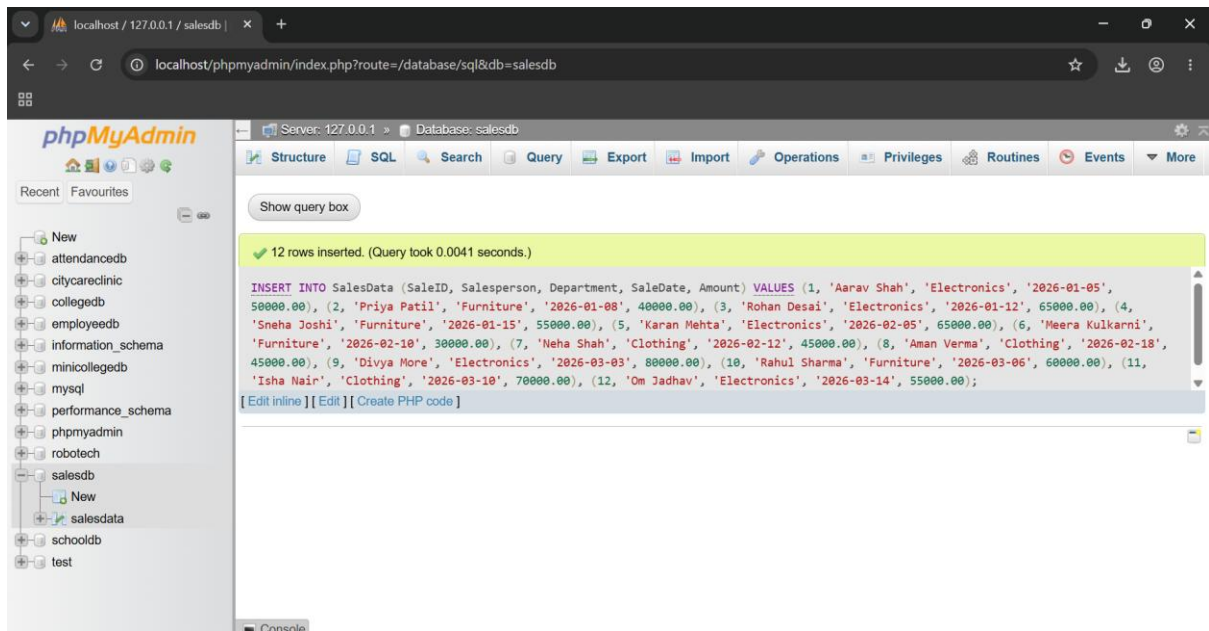
Column Name	Data Type	Constraint
SaleID	INT	PRIMARY KEY
Salesperson	VARCHAR(50)	NOT NULL
Department	VARCHAR(30)	NOT NULL
SaleDate	DATE	NOT NULL
Amount	DECIMAL(10,2)	NOT NULL

Insert at least 12 sales records belonging to three different departments.

Answer;-

```
CREATE TABLE SalesData (  
    SaleID INT PRIMARY KEY,  
    Salesperson VARCHAR(50) NOT NULL,  
    Department VARCHAR(30) NOT NULL,  
    SaleDate DATE NOT NULL,  
    Amount DECIMAL(10,2) NOT NULL  
);
```





Q2) Write a MySQL query using ROW_NUMBER() to assign a unique row number to every salesperson.

The result should be arranged by sale amount from highest to lowest.

Required Output

Column
Salesperson
Department

Amount
SalesRowNumber

Answer:-

MySQL query

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

ORDER BY Amount DESC, SaleID ASC

) AS SalesRowNumber

FROM SalesData;

Showing rows 0 - 11 (12 total, Query took 0.0005 seconds.)

```
SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER ( ORDER BY Amount DESC, SaleID ASC ) AS SalesRowNumber FROM SalesData;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

Salesperson	Department	Amount	SalesRowNumber
Divya More	Electronics	80000.00	1
Isha Nair	Clothing	70000.00	2
Rohan Desai	Electronics	65000.00	3
Karan Mehta	Electronics	65000.00	4
Rahul Sharma	Furniture	60000.00	5
Sneha Joshi	Furniture	55000.00	6
Om Jadhav	Electronics	55000.00	7
Aarav Shah	Electronics	50000.00	8
Neha Shah	Clothing	45000.00	9
Aman Verma	Clothing	45000.00	10
Priya Patil	Furniture	40000.00	11
Console\irni	Furniture	30000.00	12

Q3) Write a MySQL query using RANK() to rank salespersons according to their sale amount.

Students having the same sale amount should receive the same rank.

Required Output

Required Component
RANK()
OVER
PARTITION BY Department
ORDER BY Amount DESC

Answer:-

MySQL query

SELECT

Salesperson,

Department,

Amount,

RANK() OVER (

PARTITION BY Department

ORDER BY Amount DESC

) AS DepartmentRank

FROM SalesData

ORDER BY Department, DepartmentRank, SaleID;

The screenshot shows the phpMyAdmin interface with the following components:

- Browser:** localhost / 127.0.0.1 / salesdb
- URL:** localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=SalesData
- Server:** 127.0.0.1
- Database:** salesdb
- Table:** SalesData
- SQL Query:**

```
SELECT Salesperson, Department, Amount, RANK() OVER ( PARTITION BY Department ORDER BY Amount DESC ) AS DepartmentRank FROM SalesData
ORDER BY Department, DepartmentRank, SaleID;
```
- Results:**

Salesperson	Department	Amount	DepartmentRank
Isha Nair	Clothing	70000.00	1
Neha Shah	Clothing	45000.00	2
Aman Verma	Clothing	45000.00	2
Divya More	Electronics	80000.00	1
Rohan Desai	Electronics	65000.00	2
Karan Mehta	Electronics	65000.00	2
Om Jadhav	Electronics	55000.00	4
Aarav Shah	Electronics	50000.00	5
Rahul Sharma	Furniture	60000.00	1
Sneha Joshi	Furniture	55000.00	2
Priya Patil	Furniture	40000.00	3
Console/rni	Furniture	30000.00	4

Q5) Write a MySQL query to calculate the running total of sales according to the sale date.

Required Output

Column
SaleID
Salesperson
SaleDate
Amount
RunningTotal

Answer:-

MySQL query

SELECT

SaleID,

Salesperson,

SaleDate,

Amount,

SUM(Amount) OVER (

ORDER BY SaleDate, SaleID

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

) AS RunningTotal

FROM SalesData

ORDER BY SaleDate, SaleID;

SaleID	Salesperson	SaleDate	Amount	RunningTotal
1	Aarav Shah	2026-01-05	50000.00	50000.00
2	Priya Patil	2026-01-08	40000.00	90000.00
3	Rohan Desai	2026-01-12	65000.00	155000.00
4	Sneha Joshi	2026-01-15	55000.00	210000.00
5	Karan Mehta	2026-02-05	65000.00	275000.00
6	Meera Kulkarni	2026-02-10	30000.00	305000.00
7	Neha Shah	2026-02-12	45000.00	350000.00
8	Aman Verma	2026-02-18	45000.00	395000.00
9	Divya More	2026-03-03	80000.00	475000.00
10	Rahul Sharma	2026-03-06	60000.00	535000.00
11	Isha Nair	2026-03-10	70000.00	605000.00
12	Om Jadhav	2026-03-14	55000.00	660000.00

Q6) Write a MySQL query to display the highest-performing salesperson from each department.

The query should:

Requirement
Use ROW_NUMBER()
Partition records by department
Arrange sale amounts in descending order
Use a subquery
Display only the first-ranked salesperson from every department

Answer:-

MySQL query

SELECT

Salesperson,

Department,

Amount

FROM (

SELECT

Salesperson,

Department,

Amount,

ROW_NUMBER() OVER (

PARTITION BY Department

ORDER BY Amount DESC, SaleID ASC

) AS DepartmentRowNumber

FROM SalesData

) AS DepartmentSales

WHERE DepartmentRowNumber = 1;

The screenshot shows the phpMyAdmin web interface in a browser. The address bar indicates the URL: localhost/phpmyadmin/index.php?route=/table/sql&db=salesdb&table=SalesData. The interface includes a sidebar with a database tree on the left and a main content area on the right. The main area displays a SQL query and its results.

Server: 127.0.0.1 > Database: salesdb > Table: SalesData

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0007 seconds.)

```
SELECT Salesperson, Department, Amount FROM ( SELECT Salesperson, Department, Amount, ROW_NUMBER() OVER ( PARTITION BY Department ORDER BY Amount DESC, SaleID ASC ) AS DepartmentRowNumber FROM SalesData ) AS DepartmentSales WHERE DepartmentRowNumber = 1;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Salesperson	Department	Amount
Isha Nair	Clothing	70000.00
Divya More	Electronics	80000.00
Rahul Sharma	Furniture	60000.00

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Console Copy to clipboard Export Display chart Create view